

Introduction to Cryptography Tools

What are cryptography tools? defined as the process of concealing the contents of a message from all except those who know the key

- Symmetric : same key
- Asymmetric : different key

What is confidentiality? simply means what is private should remain private

- Cryptography can provide confidentiality through using encryption

Characterizing Cryptographic Systems

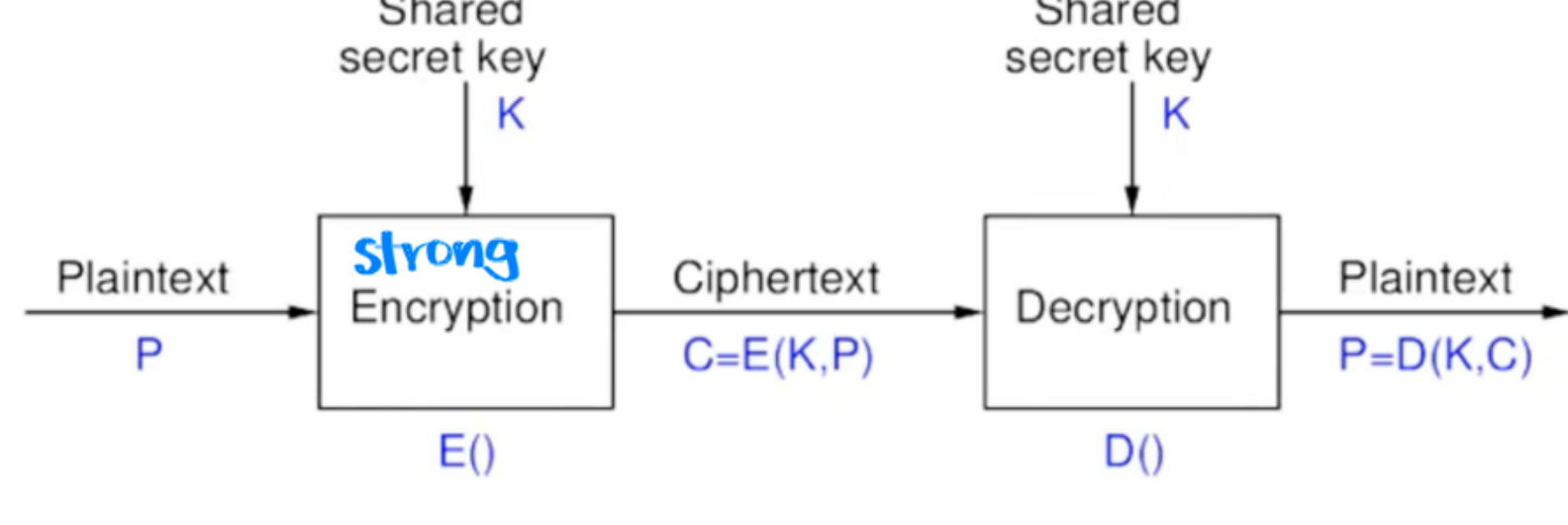
- Operations used for encryption
 - caesar cipher : shift plaintext letters K position to right
 - Substitution : replace one element in plaintext with another
 - rail-fence cipher : plaintext written in rows
 - Transposition : rearrange elements
 - Product systems : multiple stages of substitution & transposition
- Types of keys used
 - Symmetric : sender/receiver use same key
 - Public : sender/receiver use different key (asymmetric)
- Processing of plaintext
 - Block Cipher : process one block of elements at a time
 - more common
 - Stream Cipher : process input elements continuously
 - faster & use less code

Attacks

- Goal
 - Discover the plaintext (Good)
 - Discover the key (Better)
- Assumed attacker knowledge
 - Ciphertext
 - Algorithm
 - Other pairs of (plaintext, ciphertext) using same key
- Attack methods
 - Brute-force : try every possible key on ciphertext
 - Cryptanalysis : exploit characteristics of algo to deduce plaintext or key

Symmetric key encryption

- only one key (a secret key)



- Data Encryption Standard (DES) $\Rightarrow$ Key size considered INSECURE
Algorithm considered SECURE
Use 3DES

- Block size : 64 bits
- Key length : 56 bits (2^{56} keys) (64 bits, but 8 are parity)

3DES

- Three times slower than DES
- Applies DES three times

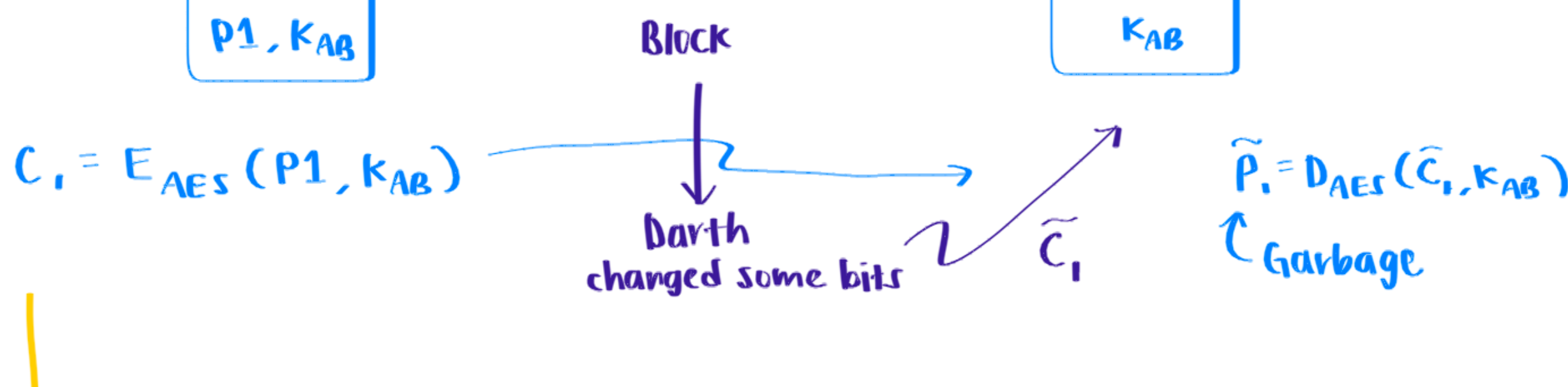
Advanced Encryption Standard (AES)

- Block size : 128
- Key length : 128, 192, 256 bits (2^{128} keys)
- Substitution-permutation network (used in HTTPS)

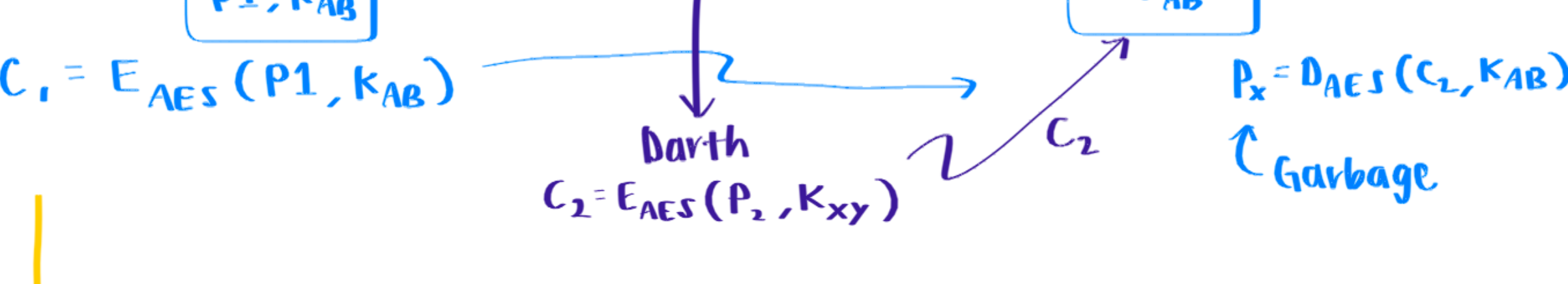
Attacks

- Brute-force : depends on key length & computer speed
- Cryptanalysis

Data Authentication



User Authentication

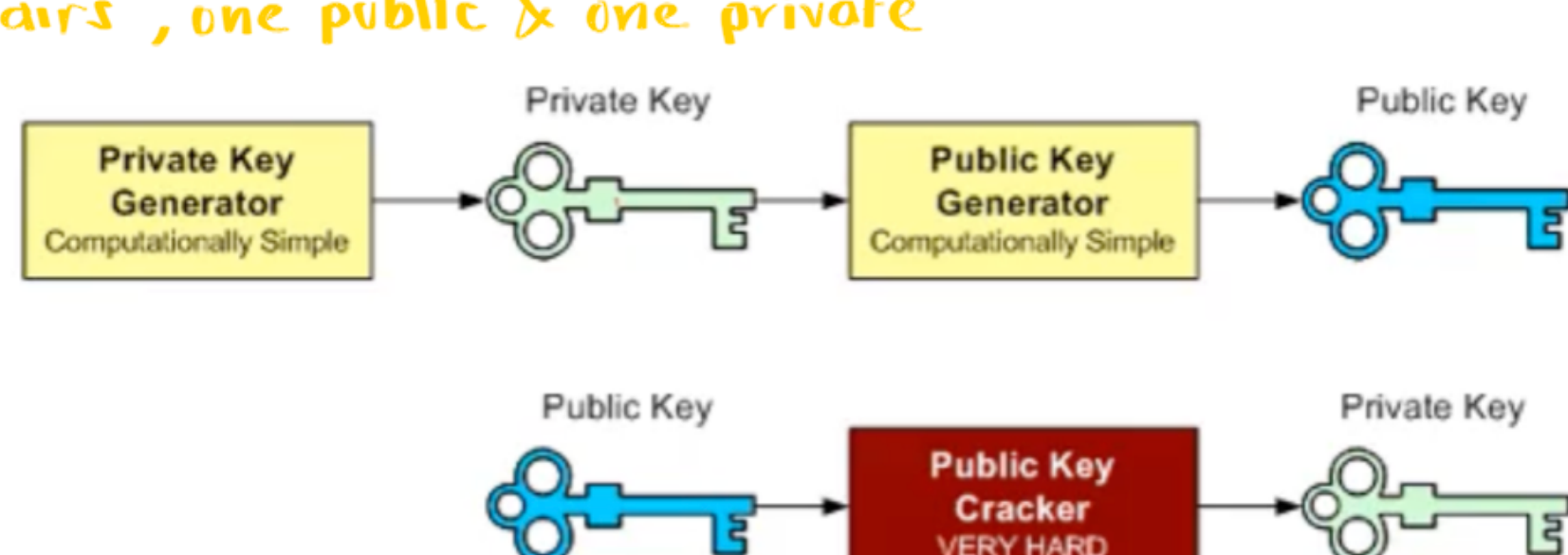


Data Confidentiality



Public Key Encryption

- Keys come in pairs, one public & one private



Examples

- Rivest, Shamir, Adleman (RSA)
- Diffie-Hellman Key exchange algorithm
- Digital Signature Standard (DSS)
- Elliptic curve cryptography (ECC)

Algorithm	Digital Signature	Symmetric Key Distribution	Encryption of Secret Keys
RSA	Yes	Yes	Yes
Diffie-Hellman	No	Yes	No
DSS	Yes	No	No
Elliptic Curve	Yes	Yes	Yes

Confidentiality

Symmetric VS Asymmetric

- Suppose 10 users need to communicate with each other
 - How many symmetric keys? 45
 - How many asymmetric keys? 20
- Public key encryption slower than symmetric due to algorithm complexity
- encrypt large data? use symmetric

Key Management / Exchange

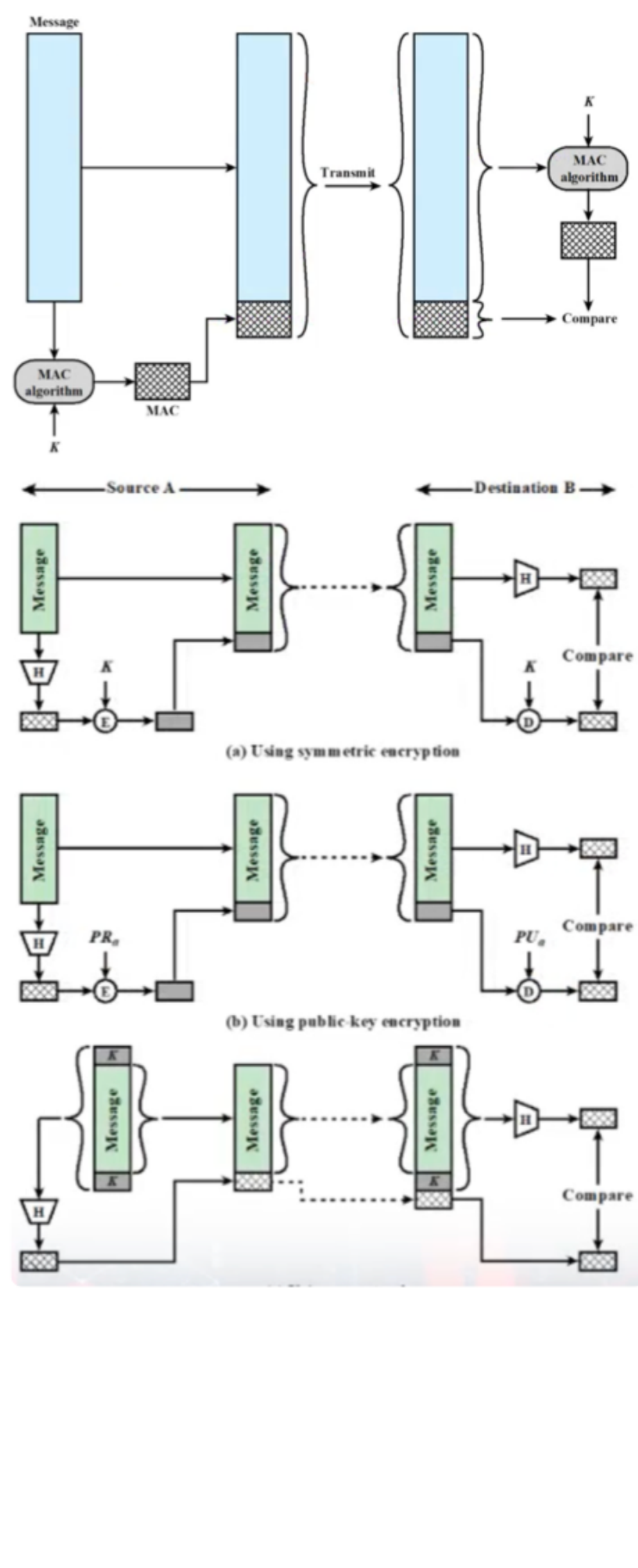
- Public key : Trusted entity / organization
- Private key : Use public-key encryption to send symmetric key

Key Generation

- Open SSL

Hashing Algorithms

- One way
- integrity
- Message Authentication Code (MAC)
 - Also called keyed hash function
 - Ex: HMAC
- HASH : applications - passwords, intrusion detection
 - does not take a key during hashing process
 - gives a fixed size hash
 - Ex: MD5 (message digest), SHA-1 (secure hashing algo), SHA-2, SHA-3
- Attack against HASH :
 - Cryptanalysis
 - Brute force



Digital Signature

- Equivalent to handwritten signature
- User Authentication
- Data Authentication (Data Integrity)
- Non-repudiation
- Applications :
 - sensitive email communications
 - financial transaction
 - software distribution

Ex: DSA

RSA - DSA

ECDSA

* All use public key encryption since only one user has a private key